Data-Quality Report

Toronto Island Ferry Tickets.csv | prepared 18 October 2026

Produced by the project's cleaning pipeline (data_processing.py). The original file is never modified.

1. Summary

The file is structurally clean and internally consistent: all 261,538 rows were retained. The quality issues that matter are semantic rather than technical: intervals with no activity are omitted, the data cover two exceptional years, and a handful of extreme single-interval spikes need verification. Overall verdict: **fit for historical demand analysis; not fit for real-time use or passenger-flow estimation.**

2. Structure and coverage

Item	Result
Format / size	CSV, 8.6 MB, 4 columns: _id, Timestamp, Redemption Count, Sales Count
Rows	261,538 read; 261,538 after cleaning
Granularity	Interval-level aggregates (one row per 15-minute interval with activity); not individual events, not cumulative
Coverage	2015-05-01 13:30:00 to 2025-12-21 22:30:00
Calendar days	3,888 in range; 3,882 with data; 6 with none
Days with no records	2015-05-02, 2015-05-03, 2018-01-04, 2021-02-21, 2021-02-22, 2025-01-26
Sales available?	Yes: Sales Count is present, so sales KPIs are enabled
Real-time suitability	None. Latest record 2025-12-21 22:30; the file is a historical extract
File order	Newest first; re-sorted chronologically by the pipeline

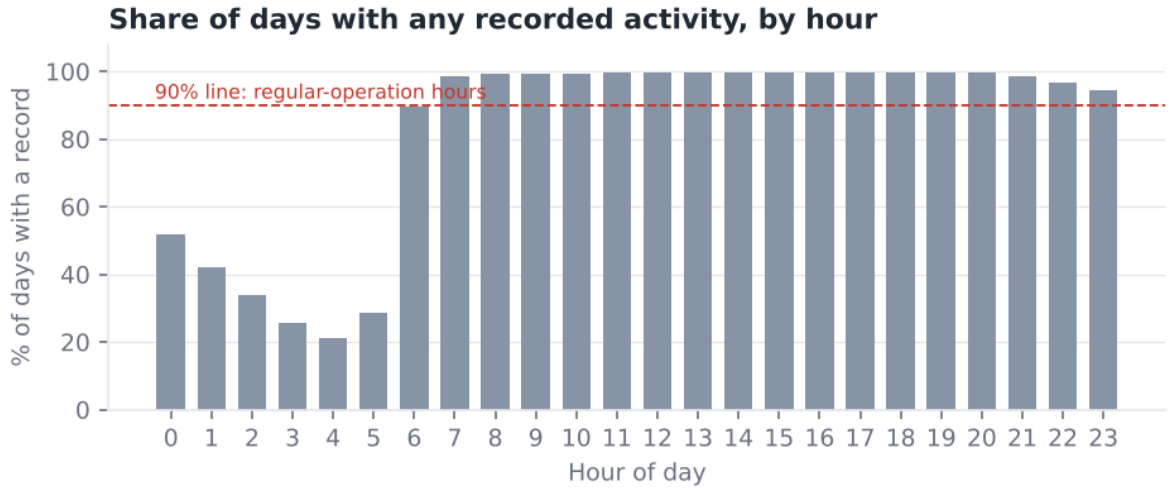
3. Validation checks and cleaning effects

Check	Result	Action
Unparseable timestamps	0	Would be dropped and counted
Missing / non-numeric counts	0	Would be dropped; never zero-filled
Negative counts	0	Would be dropped and counted
Fractional counts	0	Would be dropped and counted
Exact duplicate rows	0	Would be dropped
Repeated timestamps (different counts)	0	Would be summed per timestamp
Duplicate _id values	0	Reported only
Timestamps off the 15-minute grid	0	Would be floored for interval_start
Rows with Redemption Count = 0	27,830	Kept: genuine zero redemptions
Rows with Sales Count = 0	20,548	Kept: genuine zero sales
Rows with both counts = 0	0	None exist, so empty intervals are omitted (see section 4)
File out of chronological order	Yes	Re-sorted
Net effect	261,538 rows in, 261,538 out	No row removed or altered

Every rule is exercised by automated tests using synthetic faulty data, so the zero counts above reflect the file, not a missing check.

4. Absent intervals and completeness

Observation. No row has both counts at zero, although single-measure zeros are common. The extract therefore lists only intervals in which something was sold or redeemed. Between each day's first and last record, 63,951 of 325,489 slots are absent, a completeness of **80.3%**. Counting all 96 slots on every calendar day, completeness is 70.1%. The gap is concentrated in hours when the ferry service is not busy:



Share of days with any record, by hour. Hours 00:00-05:59 have records on only 21% to 52% of days.

Interpretation (assumption). Absent intervals most likely mean no ticket activity, not lost data. This cannot be proven from the file and should be confirmed with the data owner. For this reason the pipeline reports them as *absent*, never fills them silently, and treats them as zero only in the rolling-average calculation (user-visible option in the dashboard). There are 3,644 runs of absent intervals longer than two hours. The longest are shown below; each of the four longest contains at least one of the six days with no records, and the file does not state their cause.

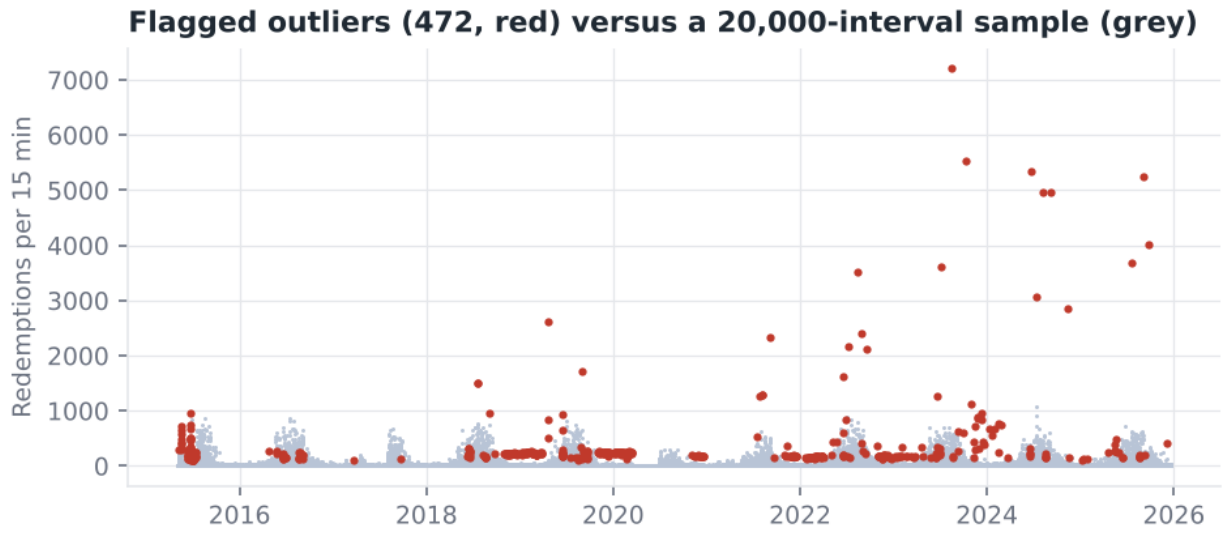
Gap start	Gap end	Absent intervals	Approx. duration
2015-05-01 16:15	2015-05-04 15:45	287	72 h
2021-02-20 18:15	2021-02-23 15:45	279	70 h
2025-01-25 23:45	2025-01-27 10:15	139	35 h
2018-01-03 23:45	2018-01-05 06:30	124	31 h
2018-01-02 07:45	2018-01-03 12:00	114	28 h
2018-01-05 07:00	2018-01-06 11:00	113	28 h
2018-02-03 11:30	2018-02-04 07:30	81	20 h
2025-12-08 16:00	2025-12-09 11:30	79	20 h

5. Time and daylight saving

Timestamps have no time zone and were kept as naive local wall-clock time; they were not treated as UTC. Evidence for local time: across 10 spring-forward days the file has 0 records in the non-existent 02:00 hour, and across 11 fall-back days there are 0 repeated timestamps. Local-time data cannot show the repeated autumn hour, so on those days the 01:00 hour may combine two real hours. The pipeline flags the 21 transition days (*dst_transition_day*) so they can be filtered or reviewed.

6. Outliers

472 intervals (0.18%) are flagged by a modified z-score (threshold 3.5) on log(1+redemptions) within month-and-hour groups, with a minimum count of 100. They hold 1.5% of all redemptions. No outlier was deleted, because peak-period demand can legitimately be extreme.

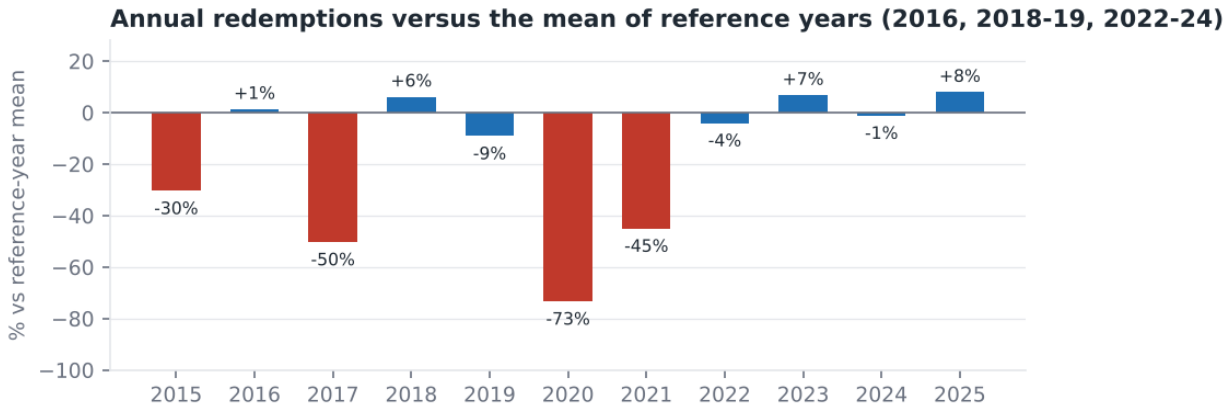


Flagged intervals (red) versus a sample of normal ones (grey).

Timestamp	Redemptions	Sales	Score
2023-08-15 20:15	7,216	7,229	7.2
2023-10-10 14:00	5,535	5,518	4.7
2024-06-19 19:00	5,341	5,342	4.6
2025-09-01 17:30	5,256	5,218	5.6
2024-08-05 22:00	4,973	4,995	5.3
2024-09-07 17:00	4,966	4,974	5.5
2025-09-24 22:30	4,020	4,015	6.7
2025-07-17 23:15	3,692	3,701	4.4

Pattern worth verifying. In each of the eight largest flagged intervals sales and redemptions differ by less than 1%. The largest (7,216 redemptions at 2023-08-15 20:15) is surrounded by intervals of 4 to 27 redemptions. Such near-identical, isolated spikes are consistent with a bulk transaction or a recording artefact; the data cannot say which. Flag counts by year: 2015: 72, 2016: 21, 2017: 2, 2018: 48, 2019: 92, 2020: 51, 2021: 25, 2022: 71, 2023: 48, 2024: 21, 2025: 21.

7. Anomalous periods



Annual redemptions relative to the mean of the six reference years (2016, 2018, 2019, 2022, 2023, 2024).

2017. May to July redemptions were 45,342, 93% below the reference-year mean for those months. **2020.** April to September were 266,124, 78% below. Both are real features of the file, not parsing errors (the days exist and the counts are consistent between sales and redemptions), but the data do not reveal whether they reflect closures, disruptions or recording problems. They should be flagged in any modelling baseline.

Sales-to-redemption ratio. 1.22 in 2015 and 1.06 in 2016, against 0.98 to 1.01 afterwards. A change of definition or process is possible; the data do not say.

Partial years. 2015 begins on 1 May and 2025 ends on 21 December; annual totals for these years are partial.

8. Limitations that affect the KPIs

- Tickets are not passengers; the sales-minus-redemptions difference is not net passenger movement.
- Per-interval and per-hour averages are computed over observed (non-absent) intervals and overstate the true time-average; the rolling average therefore uses an explicit zero-fill assumption.
- Overnight records are almost exclusively sales (zero redemptions in 98.9% of them), so off-peak ranking uses only regular-operation hours (activity on at least 90% of days).
- No route, capacity, queue, weather or event fields exist, so no analysis by route or against capacity is possible.
- The file records no publisher, licence or extraction date.

9. Recommended data improvements

- Document the definitions of Sales Count and Redemption Count, including whether empty intervals are intentionally omitted.
- Publish time-zone information or use ISO 8601 with offsets.
- Add route or terminal, vessel capacity, queue length, and event and weather flags.
- Provide a verified, regularly refreshed feed if real-time monitoring is wanted.
- Confirm the cause of the 2017 and 2020 downturns and of the extreme isolated spikes.